The Delaunay Triangulation

1. Terrain Modeling
   - height interpolation
   - Delaunay triangulations with scipy
2. Triangulations of Planar Point Sets
   - definitions
   - complexity
3. Edge Flips
   - angle optimal triangulations
   - flipping edges
   - an algorithm for optimal angle triangulations

MCS 481 Lecture 27
Computational Geometry
Jan Verschelde, 26 October 2026

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terrain modeling

Given: a set S of points in the plane, S is a sample set,
for every point we have a height measured.
Problem: represent the height at points outside S?

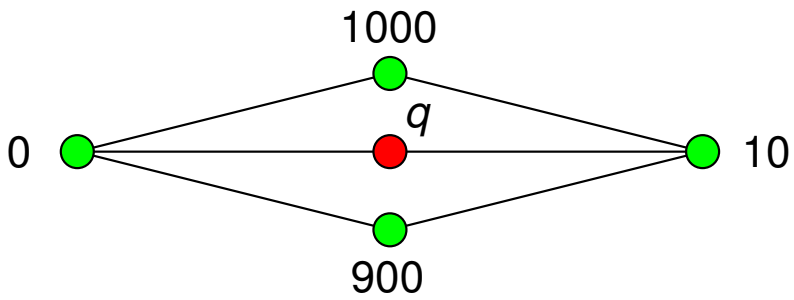
Solution: height interpolation using a triangulation of S.

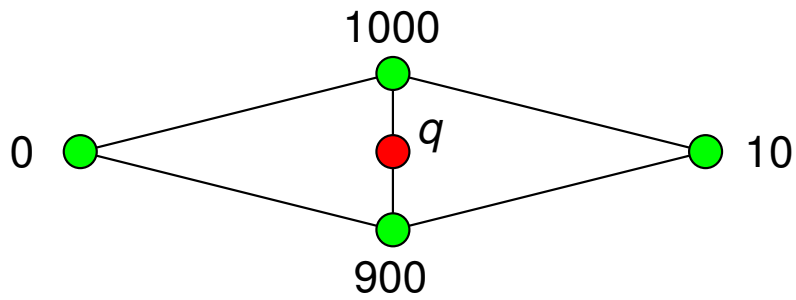
1. Determine a triangulation of S in the plane.
2. Map every triangle to 3-space, applying the heights to the points.
3. For every point inside a triangle,
the height of the piecewise linear surface models the terrain.

Question: *of the many possible triangulations,*
which one would be the best?

two triangulations

Consider two triangulations with labeled heights below:





The point q receives one of the following heights:

1. at the left, q has height $5 = (0 + 10)/2$,
2. at the right, q has height $950 = (900 + 1000)/2$.

At the left, q lies in a narrow valley (which looks wrong),
at the right, q lies on a mountain ridge (which looks better).

The optimal triangulation maximizes the minimum angle of the triangles.

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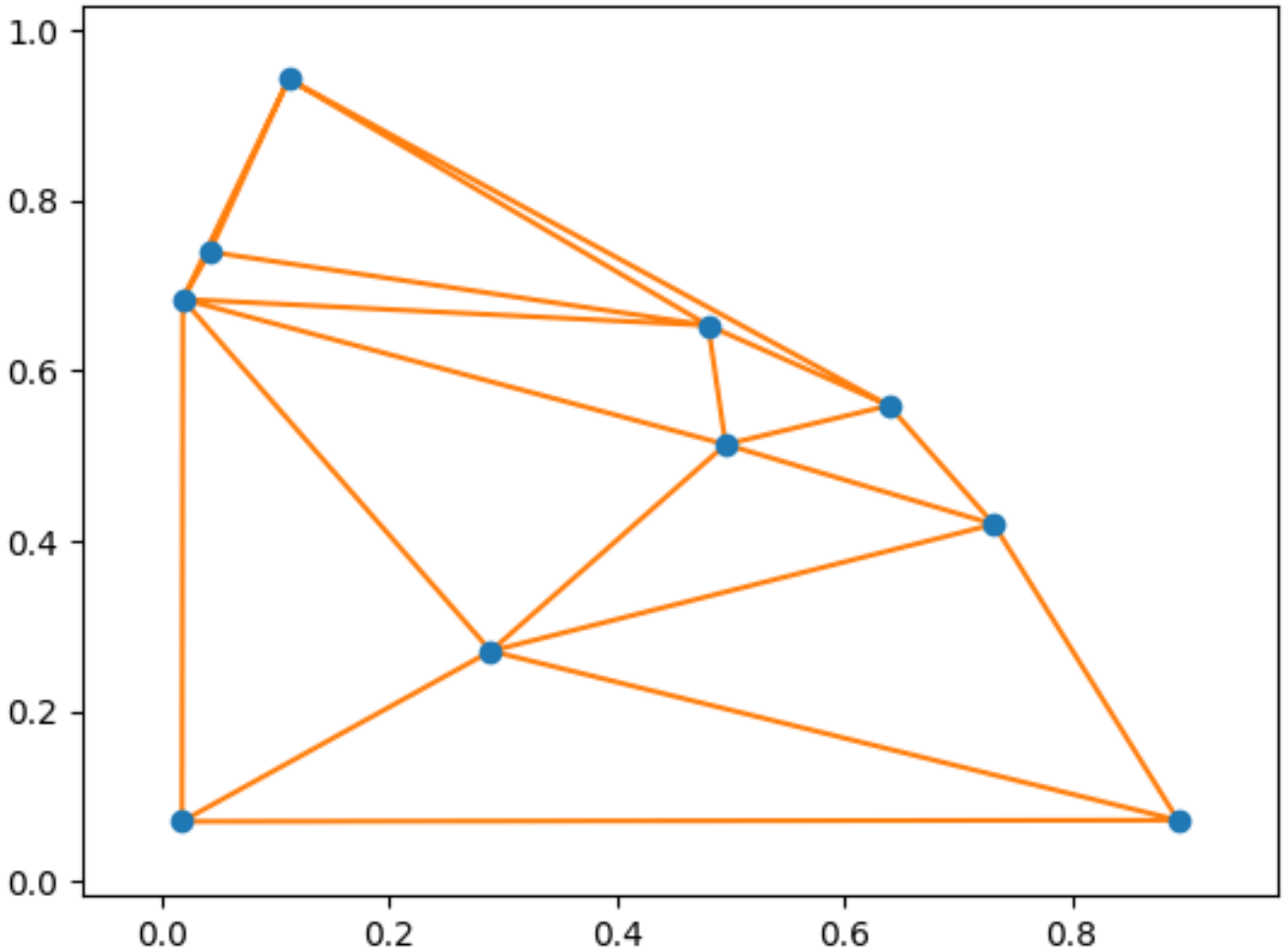
Delaunay triangulations with scipy

We use `numpy` to generate 10 random points.
The `spatial` package of `scipy` provides `Delaunay`
to construct the Delaunay triangulation and we plot with
`delaunay_plot_2d` using `matplotlib`.

```
import numpy as np
from scipy.spatial import Delaunay, delaunay_plot_2d
from matplotlib import pyplot as plt

points = np.random.rand(10, 2)
tri = Delaunay(points)
delaunay_plot_2d(tri)
plt.show()
```

the Delaunay triangulation of 10 random points



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triangulations of planar point sets

Definition (maximal planar subdivision)

A subdivision S is a *maximal planar subdivision* such that adding one edge connecting two vertices of S destroys the planarity of S.

What does destroying the planarity mean?
$\rightarrow$ Adding an edge intersects at least one existing edge.

Definition (triangulation of a point set)

Let P be a set of points in the plane. A *triangulation of the point set P* is a maximal planar subdivision which has P as its vertex set.

Observe how this differs from the convex hull problem:

- every point of the given set P must be a vertex,
- the number of edges is maximal.

all faces are triangles

Proposition (all faces are triangles)

Let P be the vertex set of a planar subdivision S.
If every face of S is a triangle, then S is maximal.

Exercise 1: write a proof for the above proposition.

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complexity

Theorem (complexity of a triangulation of a point set)

Let P be a set of n points, not all on the same line.
Let k be the number of points on the boundary of the convex hull of P.
Any triangulation of P has

1. $2n - 2 - k$ *triangles, and*
2. $3n - 3 - k$ *edges.*

If P has no inner points of its convex hull, then $k = n$ and
then any triangulation of P has

1. $n - 2$ triangles, and
2. $2n - 3$ edges.

We proved in L-7 (Theorem 3.1 in the book) that a simple polygon with n vertices has a triangulation with $n - 2$ triangles.

Euler's formula for a connected graph

Theorem (Euler's formula for a connected graph)

Let n_f, n_e, n_v respectively be the number of faces, the number of edges, and the number of vertices in a connected graph:

$$n_f - n_e + n_v = 2.$$

1. $n_v = n$, the number of points in P.
2. Let m be the number of triangles, then $n_f = m + 1$.
   With the $+1$, we count the unbounded face of the triangulation.
3. $n_e = (3m + k)/2$, as we have m triangles and every triangle has 3 edges. The unbounded face has k edges (*really?*).
   Every edge is shared by exactly 2 triangles: $(3m + k)/2$.

By Euler's formula: $n - (3m + k)/2 + m + 1 = 2$

$$\Rightarrow 2n - 3m - k + 2m + 2 = 4 \Rightarrow 2n - 2 - k = m.$$

Then, $n_e = (6n - 6 - 3k - k)/2 = 3n - 3 - k$. Q.E.D.

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the angle vector of a triangulation

Definition (the angle vector of a triangulation)

Let T be a triangulation of the point set P, $m = \#T$.
The *angle vector of the triangulation T* is $A(T) = (\alpha_1, \alpha_2, \ldots, \alpha_{3m})$,
a vector of $3m$ angles, sorted in increasing order.

By the lexicographic order on vectors,
we obtain an order on triangulations.

Let T and T' be two triangulations of the same point set P:

$$T > T' \quad \Leftrightarrow \quad A(T) >_{\text{lex}} A(T').$$

For two angle vectors α and β of size $3m$:

$$\alpha > \beta \quad \Leftrightarrow \quad \alpha_j = \beta_j, \text{ for all } j < i \text{ and } \alpha_i > \beta_i, i \leq 3m.$$

angle optimum triangulations

Definition (angle optimum triangulation)

A triangulation T of P is an *angle optimum triangulation* if $T \geq T'$, for all triangulations T' of P.

Exercise 2: Give an example of a point set for which there is more than one angle optimum triangulation.

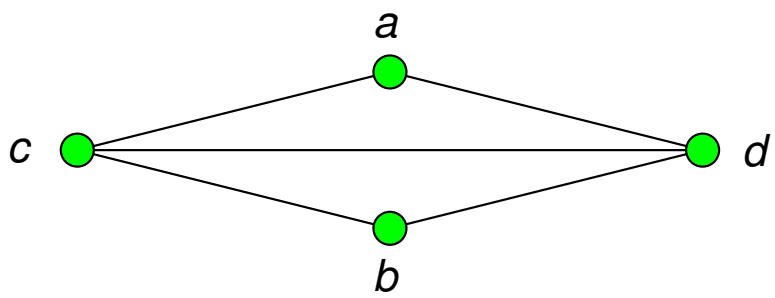
The Delaunay Triangulation

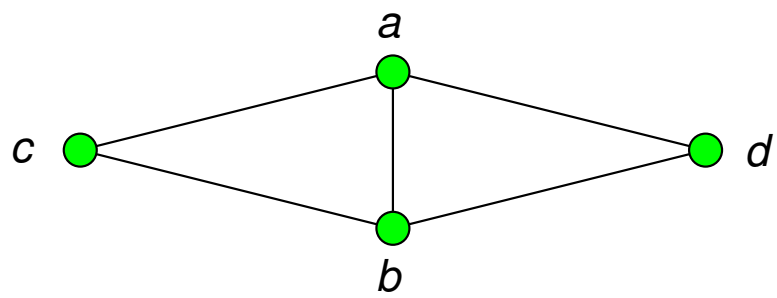
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flipping edges

Consider a convex quadrilateral spanned by vertices a, b, c, and d.
The quadrilateral has two triangulations:

1. T_1: the triangles spanned by (a, c, d) and (b, c, d), and
2. T_2: the triangles spanned by (a, b, c) and (a, b, d).





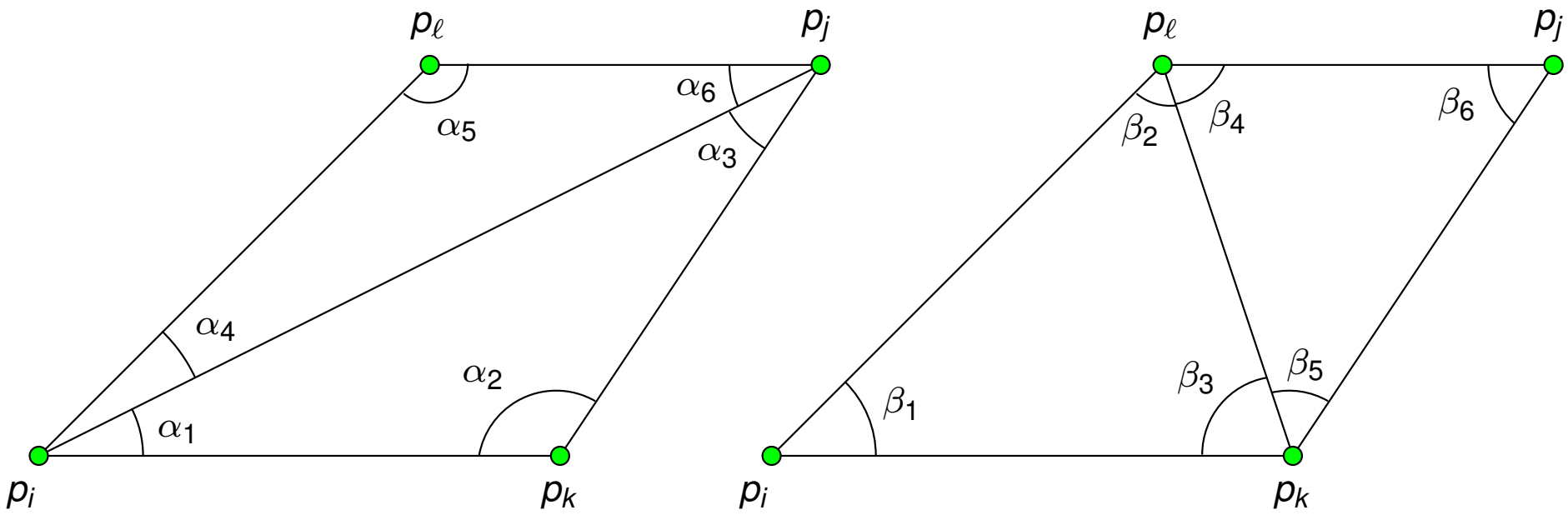
The two triangulations T_1 and T_2 are related as follows:

1. transform T_1 into T_2 by replacing (c, d) by (a, b),
2. transform T_2 into T_1 by replacing (a, b) by (c, d).

This kind of replacement is called an *edge flip*.

illegal edges

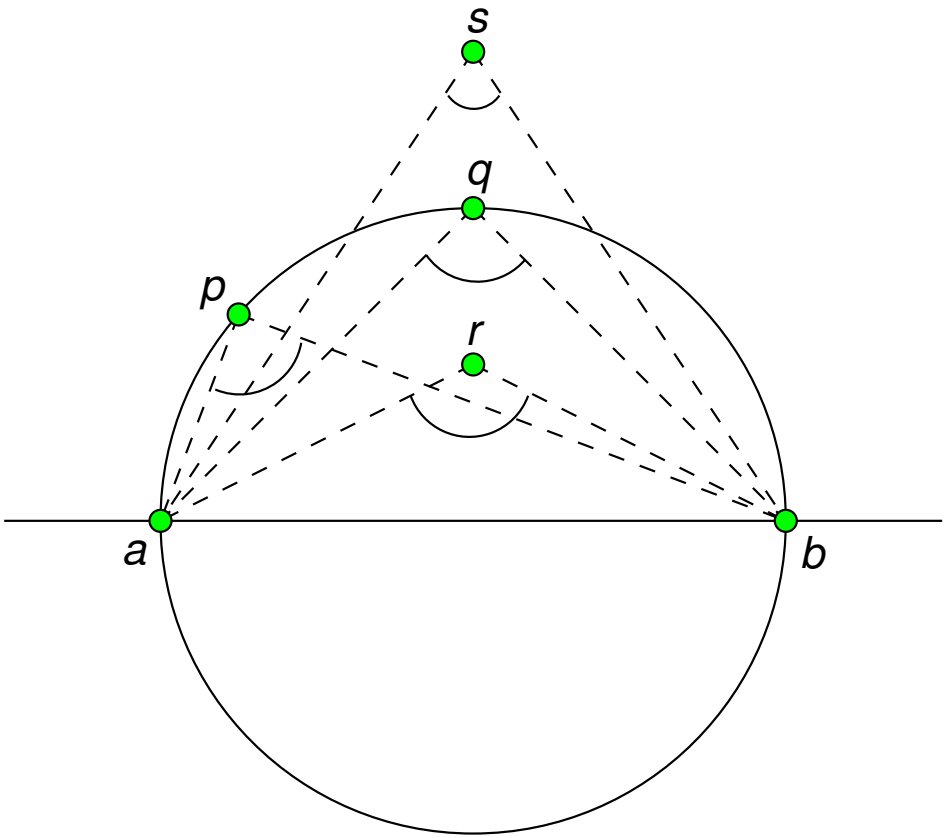
Consider the angles inside the triangles below.



Definition (illegal edge)

The edge (p_i, p_j) is an *illegal edge* if $\min_{i=1}^{6} \alpha_i < \min_{i=1}^{6} \beta_i$.

Thales circle theorem



Let $\measuredangle pqr$ denote the angle between

- the edge (p, q) and
- the edge (q, r).

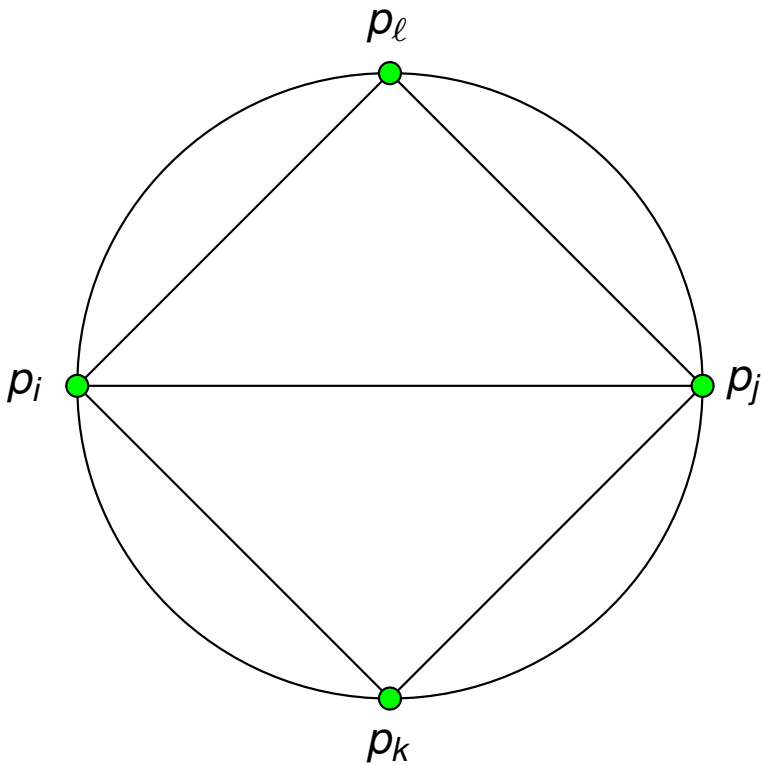
Theorem (Thales circle theorem)

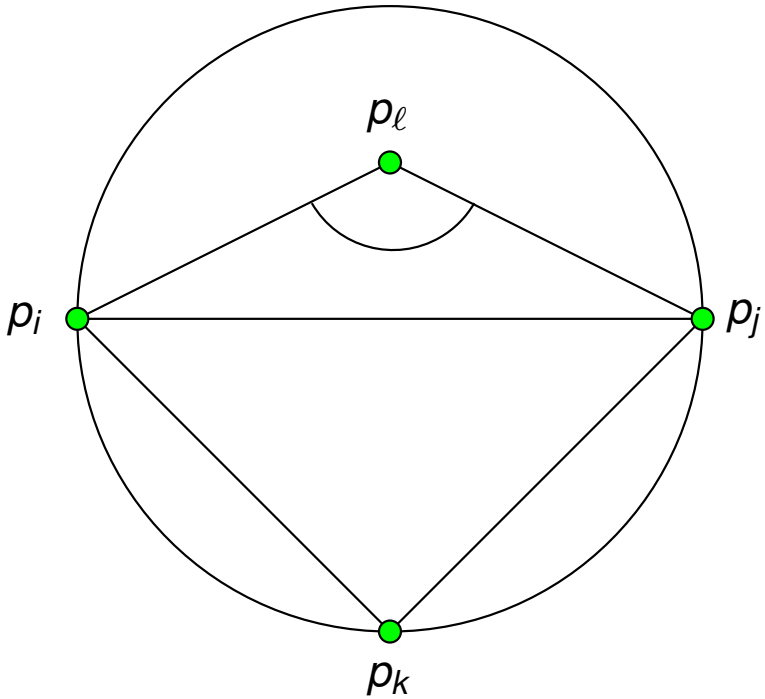
For the circle C and ℓ a line, $C \cap \ell = \{a, b\}$, $p, q \in C$,

r inside C, s outside C: $\measuredangle arb > \measuredangle apb = \dfrac{\pi}{2} = \measuredangle aqb > \measuredangle asb$.

application of the Thales circle theorem

If all points p_i, p_j, p_k, and p_ℓ are on the same circle, then both edges (p_i, p_j) and (p_k, p_ℓ) are legal.





$$\measuredangle p_i p_\ell p_j > \measuredangle p_i p_k p_j \quad \Rightarrow \quad \text{edge } (p_i, p_j) \text{ is illegal}$$

Exercise 3: Write pseudo code for a function which returns true if (p_i, p_j) is illegal, false otherwise, given p_i, p_j, p_k, and p_ℓ on input.

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an algorithm for optimal angle triangulations

A *legal triangulation* contains no illegal edges.

Algorithm LEGALTRIANGULATION(T)

Input: a triangulation T of a point set P.
Output: a legal triangulation of P.

1. while T contains an illegal edge (p_i, p_j) do
2. let (p_i, p_j, p_k) and (p_i, p_j, p_ℓ) be adjacent triangles to (p_i, p_j)
3. replace the edge (p_i, p_j) by the edge (p_k, p_ℓ)
4. return T

Why does LEGALTRIANGULATION terminate?
$\rightarrow$ With each edge flip, the angle vector of T increases.

the Delaunay triangulation

Definition (the Delaunay triangulation)

Given a point set P, *the Delaunay triangulation of P* maximizes the minimal angle over all triangulations of P.

This definition is the result of Algorithm LEGALTRIANGULATION.

In the next lecture, the dual graph of a Voronoi diagram will be called the Delaunay graph.

The `Delaunay` of the `spatial` module of `scipy` applies `qhull`, obtaining the simplices of the Delaunay triangulation of the projection of the facets of the points lifted onto a paraboloid.

summary and exercises

Given a point set P, the Delaunay triangulation of P maximizes the minimal angle over all triangulations of P.

We covered section 9.1 in the textbook.

Consider the following activities, listed below.

1. Write the solutions to exercises 1, 2, and 3.
2. Consult the CGAL documentation and example code on Delaunay triangulations.
3. Consider the exercises 1, 2, 3 in the textbook.